

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - Present	BTech	Indian Institute of Technology Kanpur	8.9/10.0
2023	XII (CBSE)	B.D Public, Patna	93.80%
2021	X (ICSE)	Don Bosco Academy, Patna	97.20%

Scholastic Achievements

- Secured **All India Rank 3734** in **Joint Entrance Examination - Advanced 2023**, among the 1,80,000+ candidates
- Rated **Expert** on **Codeforces (1607)**, ranking **628** and **815** globally in **Codeforces** round 1033 and 1003 respectively
- Received **Academic Excellence Award** for securing top **10%** rank among second-year students in the academic year 2024-2025.

Work Experience

Quant Developer Intern | Quantitative Data Insights and Technical Solutions (Jan’25 - Apr’25)

- Developed a **domain-informed custom loss** for **LSTM** with **walk-forward validation**, achieving **$R^2 > 0.55$** on predictions
- Engineered **derived technical indicators** by quantifying **cyclicality deviations** in features with **Pearson $r > 0.6$** to returns
- Added to and integrated **Python–MATLAB codebases** for **feature registration (250+)** across **500+ NSE stocks**

Multichannel Transactional Fraud Risk Scoring System | Credopay, India (Jun’25 - Jul’25)

Objective	Designing a multichannel fraud detection pipeline (POS, UPI) for anomaly detection and risk scoring
Approach	- Preprocessed (10M+ weekly) transactions, applied PCA-based compression and SHAP insights for modeling - Extracted behavioral features such as transactional velocity, geo-mismatch from 18K+ hourly rows - Employing unsupervised learning via Isolation Forest, DBSCAN, autoencoders and static rules for scoring
Results	Generating fraud risk scores (100–1000) by combining unsupervised model outputs with static rule-based scores

Research Project

Accelerated ML-Based Spatial Audio Localization | Prof. R.M. Hegde and Dr. Priyadarshini Dwivedi (May’25 - July’25)

Objective	Designing a high-accuracy pipeline for robust spatial audio localization in reverberant environments
Approach	- Sampled synthetic acoustic scenes using a 32-channel Eigenmike and advanced Pygsound simulations - Developing a workflow using Librosa and custom transforms to map RIRs to the spherical harmonic domain - Used hybrid SVM-based octant classification with CNNs and PINNs for azimuth-elevation classification
Results	Gained 22% accuracy at fixed elevation over non- Pygsound baselines; basis for upcoming research paper

Key Projects

European Option Pricing | Prof. Himanshu Yadav | Dept. of Cognitive Science, IIT Kanpur (May’25 - July’25)

Objective	Estimate fair prices of European derivatives using Markov Chain Monte Carlo (MCMC) algorithms
Approach	- Applied Metropolis-Hastings sampling over log-price trajectories using a Black-Scholes action formulation - Integrated simulated annealing for smoother convergence over complex, multi-modal posteriors - Incorporated stochastic volatility via the Heston model, tested using real NSE parameters
Impact	Validated on real NSE data, achieving less than 5% deviation under Heston and BS pricing models

QuantWeb | Team Project | Finance and Analytics Club , IITK | (May’24 - Jul’24)

- Developed an **object-oriented Python backtesting engine** with modular strategies, risk management, and **PnL** evaluation
- Developed advanced **technical analysis** tools for quantitative assessment of **10+ key market trends and indicators**
- Designed trading strategies on the volatile **Bitcoin** market using smoothing techniques and **alternating frameworks**

Stack ‘n’ Spin | Mentorship Experience | Association for Computing Activities , IITK | (May’25 - Jul’25)

- Mentored **25+ IITK** students in **Data Structures and Algorithms**, helping them build **competitive coding proficiency**
- Designing a **modular C++ CLI casino system** using **OOP principles**, enabling seamless integration of **5+ games**

Technical Skills

Programming Languages: C, C++, Python, MATLAB, JavaScript
Libraries: NumPy, pandas, Matplotlib, Yfinance Other Tools: Git, GitHub, Bash, Canva, L ^A T _E X

Relevant Courses (*Ongoing)

Fundamentals of Computing	Computer Methods for Engineers	Linear Algebra
Bayesian Modeling	Data Analytics	Data Structures and Algorithms

Extra-Curricular

- Secured a **16th** rank out of **350+ IITK** participants in the highly competitive **Optiver Trade-a-thon**
- Won **Best Summer Project** by **SNT, IITK** for developing **QuantWeb**, a live backtesting platform for the campus community
- Worked as **Secretary** of the **Chemineers IITK**, facilitating **orientation**, **farewell events**, and community-building initiatives